

29 — Saturated battery: the surplus does nothing

Proves · A battery budget past saturation: more power, identical result.

Mechanics this scenario turns on

- Past saturation extra battery power buys nothing: once ΣH is fully taken, the remaining budget has no swing left to cover.
- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\Sigma J + \Sigma L = \Sigma H$ — the sea sets the swing, the battery moves the split.

Panels described below · The invariant under test · Verified, side by side · Why this is worth a scenario rather than a unit test · The remaining 9 350.9 kW

Anything not described here — the mechanics above name the step that produced it; 00-ORIENTATION Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 10.

Scenario 01's vessel with the battery power raised from **1 260 kW to 10 000 kW**. Everything else identical, which makes this a controlled comparison against 01.

The invariant under test (INV-2)

The cascade hands the budget out row by row: $I = \min(\text{remaining}, H)$. Once the budget exceeds the **total swing ΣH** , every row takes everything it wants and the leftover is simply never used.

Beyond that point the covered band and the spinning reserve stop depending on the budget entirely.

Verified, side by side

	01 (1 260 kW)	29 (10 000 kW)
Peak shaving ΣJ	204.40	204.40
Spinning reserve ΣL	444.75	444.75
Remaining, unused	610.9	9 350.9
Battery benefit	173.66 t/yr	173.66 t/yr

Identical on every line that matters. Only the unused surplus grows.

Why this is worth a scenario rather than a unit test

The unit test `A5_SaturatedBudget_ReserveIndependentOfBudget_INV2` already asserts this at the allocation layer. What it cannot show is that the invariant survives the **whole pipeline** — that a battery eight times larger produces a byte-identical Level 1 result, the same tiers, the same CO2 and the same benefit figure.

That is a commercially meaningful statement: past saturation, a bigger battery buys nothing in this model. If someone later makes coverage scale with budget, this snapshot changes and the two numbers diverge immediately.

The remaining 9 350.9 kW

Worth reading as a diagnostic, not as waste. It says the vessel's total swing is 649.15 kW and any battery above that is oversized **for peak shaving and spinning reserve**. Other reasons to buy one — endurance, blackout recovery, harbour operation — are outside this model.

Takeaway: the calculator will honestly tell you a battery is too big. Scenario 29 makes sure it keeps doing so.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)